4

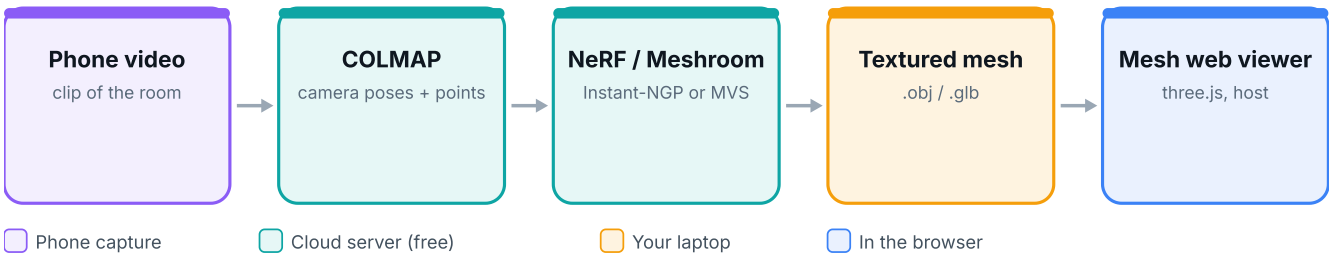
OPTION 4 · RECONSTRUCTION

Photogrammetry / NeRF Mesh

COLMAP + Instant-NGP / Meshroom — the classic textured-mesh route.

Mesh · R2 useful

In one line: the classic approach — reconstruct a **textured 3D mesh** (solid surfaces) using photogrammetry or NeRF. Great for geometry and navigation, but less photorealistic than splats for shiny / view-dependent surfaces.



Outputs a watertight textured mesh (not a splat). Good solid geometry; heavier and less sparkly than 3DGS.

How it works, step by step

1. Record a **phone video** of the room.
2. **COLMAP** recovers camera poses and a sparse point cloud.
3. **Instant-NGP (NeRF)** or **Meshroom/OpenMVS (photogrammetry)** densifies this into a surface.
4. Export a **textured mesh** (`.obj` / `.glb`).
5. Show it in a simple **three.js mesh viewer** in the browser.

Trade-offs

✓ Strengths

- Solid watertight geometry — ideal for navigation / collision
- Mature, well-documented tools
- Mesh + texture is lightweight to render
- Directly reusable for Round 2 pathfinding

✗ Weaknesses

- Less photorealistic than 3DGS (thin / shiny stuff breaks)
- Instant-NGP needs a CUDA build — heavy on a 4 GB GPU + SAC
- COLMAP slow & can fail on poor video
- More pipeline steps to maintain

Tools: COLMAP, Instant-NGP, Meshroom | Input: phone video | Output: textured mesh | Your compute: GPU (Colab better) | Cost: free

WHAT WE ACTUALLY DID

Reviewed the `From-video-to-3D-object` repo (COLMAP → Instant-NGP → MeshLab). Decided it's **superseded by Gaussian splatting** for Round 1's photorealism score, and heavy to build locally.

VERDICT FOR THE COMPETITION

Not for the Round-1 visual. But a **textured mesh is valuable for Round 2** (obstacle-aware navigation), so this is worth revisiting later alongside monomap's floor-plan.